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**Seminar in Topics in Labor Economics and the Economics of
Education**

Did Remote Work Narrow the Motherhood Penalty?

**Evidence on the Intensive Margin from Israeli Labor Force
Survey Microdata, 2017–2023**

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Abstract

Harrington and Kahn (2025) ask whether the post-pandemic rise of work from home (WFH) reduced the motherhood penalty in the United States. This paper takes that question to Israel and to the intensive margin. Using Central Bureau of Statistics (CBS) Labor Force Survey microdata on women aged 25–59 in 2017–2019 and 2021–2023, we estimate difference-in-differences (DiD) and triple-differences (DDD) models of usual weekly hours for mothers relative to childless women, with the COVID-19-driven shift to remote work as the “post” period and an Israeli-calibrated occupational WFH-exposure index as the third difference. Employed mothers’ hours did not change on average ($\text{Mother} \times \text{Post} = 0.23$, SE 0.18). The effect is instead *confined* to teleworkable work: the triple interaction is 3.22 hours per unit of exposure (SE 1.02), about 1.8 hours a week between the most and least teleworkable quartiles of occupations, while the implied change at zero exposure is if anything negative. The estimate is significant at the 1% level on the analytic occupation-clustered standard error, at the 5% level under a permutation test that reassigns exposure across occupations ($p = 0.032$), and at the 10% level under a wild cluster bootstrap ($p = 0.056$). It is unchanged under a fully saturated fixed-effects specification, survives a generalized trimming-bounds correction for selection into the estimation sample (95% confidence interval for the identified set $[1.02, 5.32]$), and is largest for mothers of children under five and smallest for mothers of teenagers. Fathers show the opposite gradient. Employment shows no average change, and the employment DDD is too imprecise to be informative. We read this as WFH access relaxing a binding time constraint for mothers in teleworkable jobs — a result about *which* mothers benefit rather than about mothers as a group.

1 Introduction

Harrington and Kahn (2025) pose a simple question for the United States: has the rise of work from home reduced the motherhood penalty in the labor market? Their answer is yes on the extensive margin and on income — a 10% rise in WFH raises mothers’ employment by roughly 0.78 percentage points relative to other women, concentrated in careers with high returns to hours and inflexible time demands, and employed mothers’ incomes rise by about 1.3% relative to other employed women. This paper asks the same question of a different country and a different margin. We use Israeli CBS Labor Force Survey microdata to test whether the post-COVID shift to remote work altered the motherhood penalty in *usual weekly work hours* among employed women. It is an extension of Harrington and Kahn’s design to another labor market and to the intensive margin, not a replication: we also estimate the employment margin, but for reasons of statistical power documented in Section 6.8 we treat it as secondary.

The shift to remote work is a rare natural experiment in workplace flexibility, which

Goldin (2014) identifies as the last major driver of the remaining gender pay gap. Temporal and locational flexibility is a good mothers pay for, in lower wages or slower promotion, and a shock that lowers the cost of supplying it should be detectable in mothers’ labor supply — for mothers already employed, most directly in how many hours they supply. Unlike the acute disruption of 2020, WFH has proven to be a persistent feature of the post-pandemic labor market (Barrero et al., 2021), which makes the question policy-relevant rather than merely historical.

The stakes are particularly high in Israel, which combines a productive, internationally integrated labor market with the OECD’s highest total fertility rate, 2.87 children per woman in 2024 against an OECD average of 1.40 (OECD, 2025). Because an unusually large share of the female workforce is, at any given time, the mother of a young child, the hours penalty is borne widely, and an intensive-margin response to flexibility has first-order consequences for aggregate labor input. Israeli practice also diverges sharply from US task-based predictions of which jobs can be done remotely: teaching, for example, returned fully to the classroom once schools reopened, whatever its task content says. That matters for how “WFH exposure” should be measured and is one of the methodological contributions below.

Our design exploits the pandemic as the moment at which remote work became a mainstream practice. In the United States, working from home rose from roughly 5% of paid workdays before 2020 to 25–30% afterward (Barrero et al., 2021). The Israeli survey began asking about remote work only in 2021, so there is no Israeli pre-period figure; in 2021–2023, between 13% and 16% of employed women aged 25–59 reported usually working from home, and 21–22% of those at work in the survey’s reference week had worked from home during it. We compare mothers and childless women across a pre-period of 2017–2019 and a post-period of 2021–2023, excluding 2020 (Section 3). A difference-in-differences on this contrast estimates the change in the hours penalty. A triple-differences model that interacts it with an occupation’s WFH exposure tests whether remote work is the *mechanism*, by asking whether the change is larger in jobs that can actually be done from home.

Our findings are as follows. Mothers’ hours did not rise on average: the DiD is 0.23 hours (SE 0.18), a reasonably precise null. The effect lives entirely in the interaction with occupational WFH exposure — 3.22 additional weekly hours per unit of exposure — while the implied change for mothers at zero exposure is if anything negative. Scaled to the regressor’s actual dispersion, that is 0.64 hours per standard deviation of exposure, or 1.8 hours between the top and bottom quartiles of occupations when the quartiles are entered directly, the same order as the 1.74-hour intensive-margin penalty itself. The estimate is 3.15 analytic standard errors from zero; because the exposure score varies across only forty occupations we also report a permutation test ($p = 0.032$) and a wild cluster bootstrap ($p = 0.056$), which place it at the 5% and 10% levels respectively. It survives a

saturated fixed-effects specification, a selection correction, two age-balance specifications, three alternative codings of the hours outcome, the exclusion of the occupations whose exposure was calibrated on post-period data, and the exclusion of 2023; it is largest for mothers of children under five; and fathers show the opposite exposure gradient. On the employment margin the DiD is a precise zero and the DDD is too imprecise to be informative.

The remainder of the paper proceeds as follows. Section 2 places the question in the literature. Section 3 describes the data, the sample, and the construction of the WFH exposure measures. Section 4 shows the hours margin and the exposure measure before any model is imposed on them. Section 5 states the hypothesis and sets out the DiD and DDD specifications, the inference, and the selection correction. Section 6 reports the results, leading with hours and closing with the employment margin. Sections 7–8 interpret the findings and state their limitations, and Section 9 concludes.

2 Literature Review

The motherhood penalty. The child penalty is one of the most robust facts in labor economics. Kleven et al. (2019b) use Danish administrative data and an event-study design around the birth of the first child to show that women’s earnings fall by roughly 20% in the long run relative to men’s, with the gap fueled by reduced work hours, exit from the labor market, and moves into family-friendly, lower-paying jobs. Kleven et al. (2019a) extend the approach across countries and find that the qualitative pattern is universal while its magnitude varies sharply, and argue that differences in gender norms, rather than in family policy, are the more likely explanation. The Danish study matters here in particular because it identifies *hours* as one of the channels through which the penalty operates — which is why an intervention that changes the time cost of combining work and care could plausibly move the intensive margin even where it does not move employment.

Temporal flexibility as the mechanism. Goldin (2014) argues that the last chapter of gender convergence turns on the structure of jobs rather than on human capital. In many occupations the return to hours is non-linear: firms pay disproportionately for long, contiguous, on-site hours, and workers who cannot supply them — disproportionately mothers — pay a penalty out of proportion to the hours they forgo. The gap would shrink if the cost of supplying a given number of hours fell. Remote work is precisely such a change: it removes the commute, allows hours to be split around care obligations, and permits a virtual presence in demanding professions. The prediction for the intensive margin is therefore direct. Mothers in occupations where the work can be done from home should supply more hours after the shift, relative to childless women in the same

occupations, with no necessary change in whether they are employed at all; mothers in occupations that cannot be done from home should show no such response. Two strands of evidence say the ingredients are real. [Mas and Pallais \(2017\)](#) show that workers, and women in particular, are willing to give up wages for the option to work from home, so the flexibility is valued; [Cortés and Pan \(2019\)](#) show that the gender gap among the skilled is concentrated where long hours are rewarded and narrows where substitutes for household time are cheap, which is the intensive-margin form of the same mechanism. Remote work may also cut the other way, through lower visibility and slower advancement, and which effect dominates is an empirical question. Goldin’s framework tells us where to look for the answer: in the gradient of the response across occupations by teleworkability, not in its average.

Does remote work narrow the gap in practice? The evidence on whether flexible arrangements realize this potential is mixed. The experimental anchor is [Bloom et al. \(2015\)](#), who find that randomly assigned home working raised work supplied without a fall in output; in a field experiment on Italian white-collar workers, [Angelici and Profeta \(2024\)](#) find that randomly assigned flexible working arrangements shifted the division of household tasks and raised women’s reported well-being. Evidence from the acute pandemic period is more ambiguous. [Adams-Prassl et al. \(2020\)](#) show that WFH-feasible jobs were better shielded from the 2020 employment and earnings shock, but that women in those jobs simultaneously absorbed a larger share of the additional childcare burden created by school closures, so the same arrangement that protected employment could degrade its intensity for mothers specifically (see also [Alon et al., 2020](#), on the macro incidence of the shock). For Israel, [Yaish et al. \(2021\)](#) document that the 2020 lockdown reallocated paid and unpaid labor within dual-earner households in gender-unequal ways. This paper adjudicates between these directions using post-acute (2021–2023) data rather than the initial-shock period that most of this literature studies.

Measuring who can work from home. [Dingel and Neiman \(2020\)](#) classify occupations by whether their tasks can be performed entirely at home, using O*NET task content, and find that 37% of US jobs can. Their classification, mapped to the two-digit ISCO-08 codes in the CBS data, is the starting point for our exposure index. As [Section 3.2](#) explains, a US task-based measure misclassifies occupations where Israeli institutional practice diverged from what the tasks would predict, and we calibrate it against realized Israeli behavior; [Madhala and Bental \(2020\)](#) provide a descriptive occupation-level baseline of WFH feasibility among Israeli workers. Any occupation-level measure captures only part of the relevant heterogeneity: [Buzaglo-Baris \(2023\)](#) shows that within-firm sorting, not occupation alone, is a first-order driver of the Israeli gender wage gap, a point [Section 8](#) returns to.

Remote work and the motherhood penalty. The paper closest to ours is [Harrington and Kahn \(2025\)](#), who show for the United States that the rise of WFH raised mothers’ employment relative to other women’s, driven by fields with high returns to hours and inflexible demands on time, and raised employed mothers’ incomes by about 1.3% relative to other employed women. That second result is an intensive-margin finding, and hours are one channel through which it could arise; our hours estimate is its closest analogue. We take the question to Israel, to hours conditional on working — with an explicit correction for the selection problem that conditioning on employment creates — and to an exposure measure calibrated to Israeli practice rather than imported from US task content. Because the exposure measure is built from occupational content rather than realized WFH status, it also avoids the reverse-causality problem inherent in treating self-reported WFH as an explanatory variable for hours.

Identifying a dose-response. Because our third difference is a continuous exposure score rather than a treated/untreated split, the design belongs to the family of difference-in-differences designs with a continuous treatment that [Callaway et al. \(2024\)](#) analyze. Two of their results shape how the estimates below are read. First, the effect of a given exposure level on the occupations that have it is identified under the familiar parallel-trends assumption, but comparing effects *across* exposure levels — which is what a single slope coefficient does — requires the stronger condition that occupations at different exposure levels would have followed the same path under any common level of exposure. Second, the linear two-way specification that economists reach for first returns a weighted average of such comparisons whose weights need not be positive or interpretable. We therefore report the triple interaction three ways: as the linear slope, under a saturated specification that absorbs every occupation-level margin except the triple interaction, and as a set of quartile contrasts whose bins are those of the descriptive dose-response figure (Section 5). Agreement across the three is the paper’s check on the functional form.

3 Data and Institutional Context

3.1 Source and sample

The data are individual-level microdata from the Israeli Central Bureau of Statistics Labor Force Survey for the survey years 2017, 2018, 2019, 2021, 2022, and 2023. The analysis sample is women aged 25–59 (CBS age-group codes 3–7). A woman is coded as a *mother* if she has any child under 17 in the household, and as childless otherwise; mothers are the treatment group and childless women the control group. *Post* equals one for survey years 2021 and later. The window is built around the pandemic’s role in making WFH a mainstream practice. The year 2020 is excluded because its arrangements

were emergency improvisation rather than the settled remote-work practice whose effect we want to measure, and because no 2020 CBS extract exists for this project at all — a constraint that also shapes the exposure index (Section 3.2).

The primary outcome is *usual weekly work hours*. The CBS records usual hours in bins; we assign each bin its median (4, 11, 18, 25.5, 32, 37, 42, 47, 54.5 and 78.5 hours) and impute the two irregular-hours codes from observed medians within the same period. Mid-distribution bins are about five hours wide, so the outcome is a bin-median approximation, and an effect smaller than one bin width — such as the headline estimate below — is best read as a shift in the share of women crossing a bin boundary rather than as a fine-grained change in hours. Hours are the only intensive-margin outcome the survey supports: the extract records how a wage is paid and by whom, but not its amount, so an earnings analysis would need a linked income source. Hours are defined for women who are employed *and* worked in the survey’s reference week.

That second restriction changes the estimand and is worth stating. In the 2017 file, respondents who were employed but absent from work in the reference week had their usual hours recorded as zero — the code CBS otherwise uses for “no usual hours” — whereas from 2018 onward the same respondents were assigned a real usual-hours value. Left uncorrected, this enters the pre-period as roughly 5,200 spurious zeros, and because absence from the reference week is more common among mothers (12.4% of employed mothers against 7.3% of childless women), those zeros fall unevenly across the comparison this paper makes. We therefore define hours on the reference-week workers in *every* year. This costs about 10% of each year’s employed sample, makes the hours population identical across years, and narrows the estimand to usual hours among women who were working rather than among all women who held a job; Section 8 returns to what this does and does not fix.

The secondary outcome is a binary employment indicator, equal to one if the respondent reports being employed and zero otherwise (unemployed and not-in-labor-force are both zero).

All regressions include the same set of categorical controls: marital status, religion, age group, district of residence, and highest educational attainment (six groups). Age enters as the categorical CBS age-group code because the extract contains no continuous age or birth-year variable. CBS survey weights are deliberately not applied in any outcome regression; all reported estimates are unweighted estimates on the analysis sample rather than population-representative statistics (Section 8).

3.2 Constructing WFH exposure

Since realized WFH status is itself a labor-market outcome — jointly determined with employment and hours — causal designs need a measure of WFH *exposure* that is not

chosen by the individual whose hours are being modeled. The mechanism test therefore requires an occupation-level measure of how exposed each job is to remote work. We build it in three steps, each producing a measure that is reported.

External. The [Dingel and Neiman \(2020\)](#) teleworkability score, via an O*NET/SOC-to-ISCO crosswalk, at the two-digit ISCO-08 level. This is pre-period by construction and immune to Israel’s own COVID-era behavior, but it is a US task-based measure and misclassifies occupations where Israeli practice diverged. Teaching (ISCO 23) is the clearest case: it scores 0.966 on the external index, yet Israeli schools, closed nationally on 14 March 2020 and reopened in stages from 3 May 2020 ([Somekh et al., 2021](#)), then closed again in the September 2020 and January–February 2021 lockdowns ([Hale et al., 2021](#)), were fully in person throughout the 2022–23 survey years the calibration draws on, and teachers did not work from home. One caveat on this measure belongs here rather than in a footnote: the crosswalk that maps the US occupation codes onto ISCO-08 is not documented in the replication materials, so a reader can check the forty resulting scores (Appendix Table A1) but not rebuild them from the source.

Calibrated (primary). We correct the external score only where the gap between it and the realized 2022–23 Israeli WFH share is large (greater than 0.5) *and* statistically distinguishable from sampling noise, by a one-sided test at the 95% level using a standard error clustered by individual. Ten of the 40 two-digit occupations are swapped to their realized Israeli values, and the corrections are concrete and checkable: teaching falls from 0.966 to 0.069, clerical support (ISCO 41) from 1.000 to 0.098, while ICT professionals (ISCO 25) go from 1.000 to 0.360. Because the swap test draws on 2022–23 realized data, which sit inside the post-period, the calibrated measure is a documented compromise rather than a fully pre-treatment index. The realized shares are computed on men and women together, so that an occupation’s score is not built from exactly the people who enter the regression; Section 6.4 sets out what the construction does and does not put at risk.

How well does the score the headline is estimated on predict where remote work actually happened? Across the forty occupations, on men and women pooled over 2021–23, the calibrated score has a correlation of 0.53 with the share who worked from home in the survey reference week and an occupation-size-weighted slope of 0.40 (SE 0.09); against the share who *usually* work from home, the item the calibration targets, the correlation is 0.48 and the slope 0.26 (SE 0.05). Both are positive and precise, so the score is relevant. But the raw external index does better on both measures (correlations 0.74 and 0.59), and the reason is instructive: swapping ten occupations onto realized shares while leaving thirty on task-based probabilities mixes two scales, so an unswapped occupation whose external score overstates its remote work by just under the swap threshold now ranks above a swapped occupation whose realized share is actually higher. The calibration’s contribution is therefore not a better cross-sectional fit; it is the removal of the

ten grossly mis-scored occupations from the treated tail, which is exactly the variation the unswapped-occupations check of Section 6.4 isolates. The forty scores are listed in Appendix Table A1 and plotted in Figure A1.

Realized. The realized Israeli WFH share by occupation, anchored to 2021 with a minimum of 200 observations per occupation. This is post-treatment by construction and is reported as a robustness check. The original research plan anchored it to 2020; with no 2020 extract it is anchored to 2021 instead, a disclosed deviation whose downside — by 2021 occupations had had a year to adapt, and unevenly — Section 8 discusses.

A fourth measure serves two roles. For the employment DDD, occupation is undefined for the non-employed, so an occupation-level regressor would condition the outcome on itself; that specification instead uses a pre-period (2017–2019) shift-share index by demographic cell (sex, age group, education, district, marital status, religion, and continent of birth — 8,884 cells), equal to the weighted mean of the calibrated occupation score over the cell’s pre-period occupational composition and defined for employed and non-employed women alike. The same index, because it is observable for the non-employed, stratifies the selection correction of Section 5.5. Its scale is not that of the occupation-level index, and we never compare coefficients across the two as if they were on a common scale.

4 Descriptive Statistics

This section sets out what the analysis sample looks like before any model is imposed on it: the summary table, then the primary margin, then the exposure measure that identifies the triple difference. Nothing here is adjusted for the controls listed in Section 3, and nothing here carries an identifying assumption. All standard errors and confidence intervals are clustered by individual, as they are in Section 6, because the survey is a rotating panel (Section 5.4); the point estimates are ordinary cell arithmetic, but the intervals around them are not.

4.1 The sample in summary

Table 1 describes the sample on either side of the treatment/control split. Two features matter for what follows. Mothers work *fewer* hours than childless women when they work (38.68 versus 40.06) while being slightly *more* likely to be employed (0.7828 versus 0.7574), so the *level* of the penalty is visible in the raw data on the intensive margin; whether that level changed after 2021 — the question this paper actually asks — is not something the table can show. And the two groups are imbalanced in ways the controls must absorb: mothers are the better-educated group (47.97% with an academic degree against 40.24%), overwhelmingly married (84.57% against 46.55%), and about 0.44 of an

age-group code younger — an imbalance Section 6.4 returns to because it tracks WFH exposure.

4.2 The raw hours gap

Before any model, the primary outcome in four cell means. Mothers work 38.54 hours per week before 2021 and 38.84 after, a gain of 0.31 hours. Childless women move from 40.04 to 40.08, a gain of 0.04. The difference between those two changes is 0.266 hours, and it is the entire difference-in-differences of equation (1) computed without a single control variable.

This matters for how the later estimate should be read. The regression DiD of Table 2 is 0.2280 (SE 0.1809), within four hundredths of an hour of the raw figure and with an almost identical standard error. What the controls do not do, in other words, is manufacture or conceal an average effect: the near-absence of one is already present in the arithmetic of four averages, and neither version is distinguishable from zero. Mothers remain well below childless women in both periods. The average mother’s hours are not what moves in this paper; Section 4.4 shows what does.

4.3 Hours year by year

Pooling the pre-period into a single average hides whether the two groups were on a common trajectory before 2021, so Figure 1 unpools it. The lower panel plots the raw mother-minus-non-mother hours gap in each survey year. The pre-period gap is -1.37 hours in 2017, -1.48 in 2018 and -1.67 in 2019: three years within about a third of an hour of each other, and each within the others’ confidence bounds. Whatever else is true of this design, the groups were not diverging before treatment; this is the descriptive counterpart of the formal test in Figure 3. The same panel shows the post-period path: -1.37 in 2021, -1.53 in 2022 and -0.80 in 2023, the smallest value in the sample. The movement is not monotone and the three post-period years are not individually distinguishable, so it is the event study of Section 6.1, not this raw series, that carries the timing reading of Section 7.

4.4 Exposure and the commuting channel

The triple difference asks whether hours respond where work can be done from home. Figure 2 puts that question to the data directly, by computing the same difference-in-differences as Section 4.2 separately within each quartile of occupational WFH exposure. The answer is not that the response is larger at the top; it is that there is no response anywhere else. The bottom three quartiles return -0.32 , 0.27 and -0.02 hours, none of them distinguishable from zero. The top quartile returns 1.55 hours (95% CI $[0.74, 2.36]$).

Table 1: Summary statistics: women aged 25–59, by mother status

	Childless women	Mothers	Difference (SE)
<i>Panel A: outcomes</i>			
Observations	133,646	239,095	
Employment rate	0.7574	0.7828	0.0254 (0.0031)
Usual weekly hours (ref.-week workers), mean	40.06	38.68	−1.38 (0.09)
standard deviation	[12.12]	[11.62]	
observations with usual hours	93,982	164,193	
<i>Panel B: characteristics</i>			
Age-group code (CBS 3–7), mean	5.40	4.96	−0.44 (0.01)
Arab (%)	17.73	17.78	0.05 (0.33)
<i>Marital status (%)</i>			
Married	46.55	84.57	38.02 (0.35)
Married, living separately	3.58	0.84	−2.74 (0.11)
Divorced	13.13	7.70	−5.42 (0.24)
Widowed	2.75	0.99	−1.76 (0.11)
Single, never married	34.00	5.90	−28.10 (0.31)
<i>Religion of household head (%)</i>			
Jewish	73.08	79.50	6.42 (0.36)
Christian	7.12	2.23	−4.89 (0.18)
Muslim	13.78	14.89	1.11 (0.30)
Druze	1.88	1.64	−0.24 (0.12)
Other	4.15	1.75	−2.40 (0.14)
<i>District of residence (%)</i>			
Jerusalem	8.76	12.31	3.56 (0.26)
North	17.52	16.58	−0.94 (0.32)
Haifa	14.38	11.92	−2.45 (0.28)
Center	23.99	26.52	2.53 (0.36)
Tel Aviv	21.55	14.90	−6.65 (0.33)
South	11.66	12.37	0.72 (0.27)
Judea and Samaria	2.15	5.39	3.24 (0.15)
<i>Education (%)</i>			
Below high school	11.37	6.63	−4.74 (0.23)
High school, no matriculation	12.70	11.95	−0.74 (0.24)
Matriculation (Bagrut)	22.73	18.82	−3.90 (0.29)
Post-secondary, non-academic	11.50	13.68	2.18 (0.24)
Academic degree (BA/MA/PhD)	40.24	47.97	7.73 (0.39)
Other / no certificate	1.47	0.95	−0.52 (0.07)

Israeli CBS Labor Force Survey, survey years 2017–2019 and 2021–2023, women aged 25–59. A woman is a mother if she has any child under 17 in the household. Weekly hours are conditional on working in the reference week (Section 3); the standard deviation is in brackets and the hours rows carry their own observation count. Within each categorical block, percentages sum to 100 in each column. The difference column is mothers minus childless women, with a standard error clustered by individual in parentheses. All figures are *unweighted* counts on the analysis sample and are not population-representative (Section 8). The Arab share is the CBS population-group code, the variable used for the subgroup analysis; religion is a separate CBS item recorded for the household head. Level labels follow the CBS codebook that accompanies the extract. Marital status, religion, district, age group and education are the five controls in every regression.

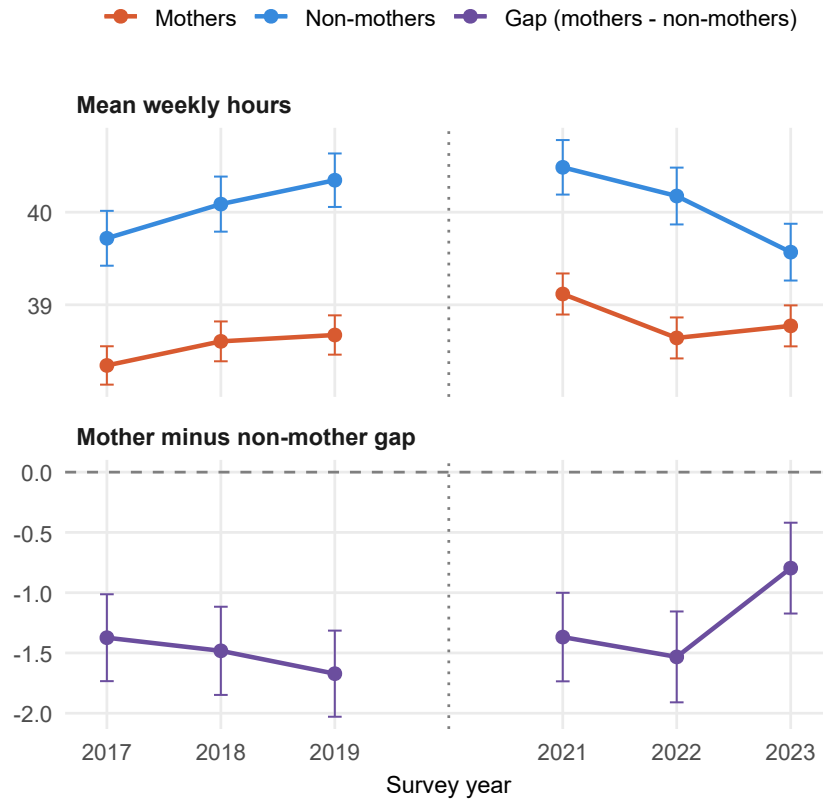


Figure 1: Weekly hours by survey year: levels and the motherhood gap
Notes: Women aged 25–59 who worked in the reference week, unweighted means with no controls. The upper panel plots each group’s mean weekly hours; the lower panel plots their difference. Vertical bars are 95% confidence intervals clustered by individual. The dotted rule marks the excluded 2020 survey year, and the series break there rather than interpolating across it. The three pre-period gaps are the raw counterpart of the insignificant $\text{Mother} \times \text{year}$ coefficients in Figure 3.

Two things follow. First, this is why Section 4.2’s average is near zero: it is a weighted average of three nulls and one substantial positive difference, and the three nulls dominate by construction. The decomposition is approximate rather than exact — these quartiles additionally require a matched occupation code, so they rest on 251,968 rows against Section 4.2’s 258,175, and the quartile-weighted average is 0.29 rather than 0.27. Second, the dose-response that the DDD coefficient of 3.224 summarises as a single slope is visible without any regression — but it is emphatically not linear in exposure, and the identifying contrast is a comparison of the most teleworkable quartile against the rest rather than a smooth gradient. The quartile boundaries make the same point: the bottom three quartiles span exposure values of 0.00 to 0.30, while the top quartile spans 0.30 to 0.75. Most of the dispersion in the regressor lives in one tail, which is worth bearing in mind when reading a coefficient expressed per unit of exposure.

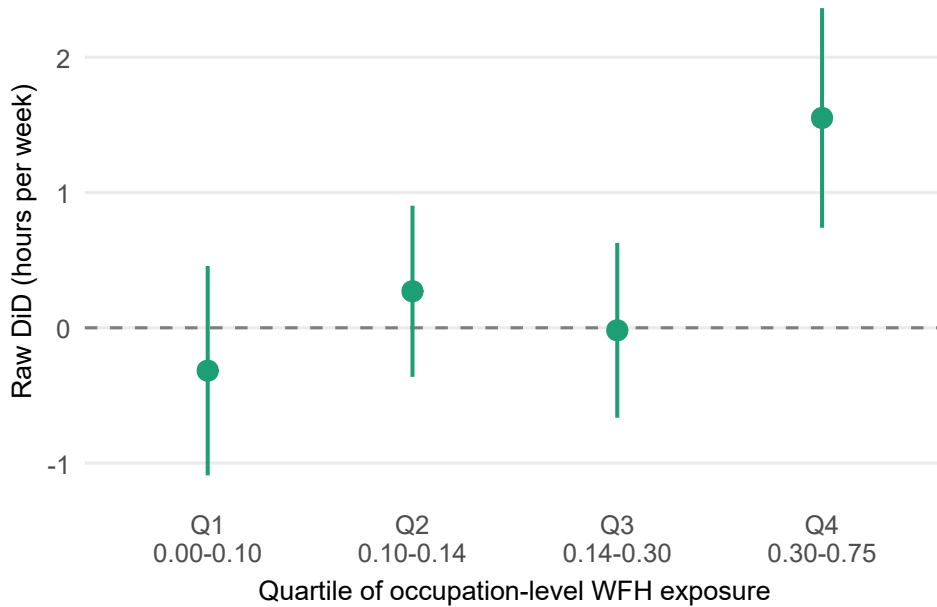


Figure 2: Raw hours difference-in-differences by quartile of WFH exposure

Notes: Women aged 25–59 who worked in the reference week and carry a matched occupation code. Each point is the (mothers post – mothers pre) – (childless post – childless pre) difference within one exposure quartile, from unweighted cell means with no controls; bars are 95% confidence intervals clustered by individual. Quartiles are of the *occupation-level calibrated* exposure measure, the same regressor the hours DDD uses. Breakpoints are computed on pre-period rows and applied to all years, and each quartile’s exposure range is printed beneath its tick: the bottom three quartiles together span exposure 0.00 to 0.30 while the top spans 0.30 to 0.75.

Commuting patterns are consistent with this channel. Among women aged 25–59 with a recorded locality of work, mothers were the *more* mobile group in every year — working outside their locality of residence 48.2% of the time in 2019 against 46.1% of childless women — and their commuting was the less disturbed by the pandemic: the share fell 1.5

percentage points for mothers in 2021 against 3.9 for childless women, a difference of 2.39 points (SE 1.17), with both series back at their 2019 levels by 2023. That fits a mechanism in which remote work relaxes a time constraint for women who were already commuting rather than drawing a previously home-bound group into work. It is descriptive evidence on the mechanism rather than a test of it; no specification in this paper uses locality of work as an outcome.

5 Empirical Strategy

5.1 Hypothesis

The flexibility framework of Goldin (2014), as set out in Section 2, yields one testable hypothesis. **H1:** after 2021, mothers' usual hours conditional on working rise relative to childless women's in proportion to their occupation's WFH exposure, with no response where the work cannot be done from home. The average change in mothers' hours is not the object of interest; the exposure gradient is. We also estimate the employment margin for completeness (Section 6.8).

5.2 Difference-in-differences

For outcome Y_{it} (weekly hours conditional on working that week, or employment) we estimate

$$Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}, \quad (1)$$

where X_{it} is the control vector described above. β_3 is the DiD estimator: the differential change in the outcome for mothers relative to childless women after the shift to remote work. For hours, a positive β_3 indicates a narrowing of the penalty on the intensive margin.

5.3 Triple differences

To test whether the narrowing operates through remote work, we interact the DiD with occupational WFH exposure:

$$\begin{aligned} Y_{it} = & \alpha + \delta_1 \text{Mother}_i + \delta_2 \text{Post}_t + \delta_3 \text{WFH_Exposure}_{j(i)} + \delta_4 (\text{Mother}_i \times \text{Post}_t) \\ & + \delta_5 (\text{Mother}_i \times \text{WFH_Exposure}_{j(i)}) + \delta_6 (\text{Post}_t \times \text{WFH_Exposure}_{j(i)}) \\ & + \delta_7 (\text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)}) + X'_{it} \gamma + \varepsilon_{it}, \end{aligned} \quad (2)$$

where $j(i)$ is individual i 's occupation. The coefficient of interest, δ_7 , asks whether the post-shift change for mothers relative to non-mothers is larger in more exposed occupa-

tions. The lower-order terms do specific work. δ_6 absorbs any post-2021 change in hours in teleworkable occupations that is common to mothers and childless women alike — the general remote-work recovery — so that δ_7 is not contaminated by it. δ_5 absorbs the pre-existing level of the motherhood penalty across occupations, so that δ_7 is a change and not a level. And δ_4 is the post-shift change for mothers at *zero* exposure, not an average effect. Following [Olden and Møen \(2022\)](#), the DDD does not require two parallel-trends assumptions but a single one: that the mother/non-mother gap in high- versus low-exposure jobs would have evolved in parallel absent the shift to remote work.

For the hours DDD, WFH_Exposure_j is the calibrated occupation-level score of [Section 3.2](#) and the sample is employed women matched to a two-digit occupation. For the employment DDD, WFH_Exposure is the cell-based shift-share index, the sample is all women, employed or not, and a $\text{Mother} \times \text{age-group}$ interaction is added to correct a documented pre-period age imbalance between mothers and non-mothers.

Equation (2) makes each lower-order occupation-level difference linear in exposure. With only forty occupations that restriction is unnecessary, so we estimate two relaxations of it alongside the pooled model. The *saturated* specification replaces every lower-order term with fixed effects,

$$Y_{it} = \delta_7(\text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)}) + X'_{it}\gamma + \lambda_{j(i),t} + \mu_{j(i),\text{Mother}_i} + \nu_{\text{Mother}_i,t} + \varepsilon_{it}, \quad (3)$$

where λ_{jt} are occupation-by-survey-year effects (absorbing δ_2 , δ_3 , δ_6 and any occupation-specific year shock), $\mu_{j,m}$ are occupation-by-mother effects (absorbing δ_1 and δ_5), and $\nu_{m,t}$ are mother-by-year effects (absorbing δ_4). Only the triple interaction carries a functional form. The *binned* specification keeps the pooled model’s structure but replaces $\text{WFH_Exposure}_{j(i)}$ with indicators for the quartile of exposure the occupation falls in — the same pre-period quartiles as the dose-response figure of [Section 4.4](#) — so that the top-versus-bottom contrast is reported as a coefficient with a standard error rather than read off a figure. For each DDD we also report the minimum detectable effect, $\text{MDE} = \text{SE} \times (z_{0.975} + z_{0.80}) = 2.8016 \times \text{SE}$, so that a null coefficient can be read as “genuinely null” or “underpowered.”

5.4 Inference

The survey is a rotating panel: the hours estimation sample of 258,175 observations comes from only 64,602 distinct respondents — about four observations each, and two to three within a single survey year, with 96% of rows belonging to someone who appears more than once. Treating those observations as independent would understate standard errors by a factor of roughly 1.8, so equation (1) and every descriptive interval in [Section 4](#) are clustered by individual. What the panel structure does *not* support is within-person identification: only 1.9% of respondents are re-interviewed across the pre/post boundary,

so an individual fixed-effects design was considered and rejected.

For the hours DDD the regressor of interest is constant within a two-digit occupation, and standard errors are clustered at that level — individual-level clustering would understate them. That leaves forty clusters. The reported p -values use `fixest`’s default small-sample convention, a t reference distribution with $G - 1 = 39$ degrees of freedom and a cluster-count adjustment to the variance. Forty is few enough that we do not rest on that approximation. For the headline coefficient and for the pre-period coefficients of the triple-difference event study we also report wild cluster bootstrap p -values, with Rademacher weights, the null imposed and 9,999 draws (Roodman et al., 2019), computed with the `fwildclusterboot` package (Fischer and Roodman, 2021). The *joint* pre-period test remains the analytic Wald F : the joint bootstrap is not available in that implementation without a Julia toolchain, so we report the bootstrap for each pre-period coefficient and for their sum instead. Section 5.6 adds a permutation test that relies on no asymptotic approximation at all. Clustering two ways, by individual and by occupation, leaves the headline standard error unchanged (Section 6.4). The employment DDD is clustered on the age-by-education-by-district cell (about 210 clusters).

5.5 Selection into employment: Lee bounds

Hours are observed only for women who are working, and whether a woman is working is itself an outcome of the same treatment. If remote work differentially pulls marginal mothers into work after 2021, the post-period sample of working mothers is compositionally different from the pre-period one for reasons unrelated to hours choices, biasing β_3 in the hours equation in a direction that is not signable a priori. We address this with trimming bounds in the spirit of Lee (2009), adapted to the two-by-two design. Let $s_{ab} = \Pr(\text{hours observed} \mid \text{Mother} = a, \text{Post} = b)$ — that is, the probability of being employed *and* working in the reference week, which is the sample the outcome is measured on rather than the employed population as a whole. The counterfactual below assumes parallel trends *in selection*. For post-period mothers it is $s_{11}^* = s_{10} + (s_{01} - s_{00})$. If the observed s_{11} exceeds s_{11}^* , the excess share $p = 1 - s_{11}^*/s_{11}$ is trimmed from the (Mother = 1, Post = 1) cell’s hours distribution — from the top to obtain a lower bound on β_3 , from the bottom to obtain an upper bound — and equation (1) is re-estimated on each trimmed sample. Trimming from a single tail assumes monotone selection: that remote work weakly increases, and never decreases, a mother’s probability of being observed with hours.

For the DDD we generalize this construction. The selection counterfactual is computed separately within each quartile of the cell-based exposure index (which, unlike the occupation-level index, is observable for the non-employed), each quartile’s post-period mothers are trimmed by that quartile’s own excess share, the trimmed subsets are recom-

bined, and equation (2) is refit with the occupation-level regressor. This stratification assumes that selection into observed hours depends on exposure only through the cell-index quartile; it is an approximation, since the outcome equation uses the occupation-level index, and the bound is valid to the extent that assumption holds. Because the parameter is only partially identified, we report, alongside each bound’s own cluster-robust confidence interval, the Imbens and Manski (2004) confidence interval for the identified set, which accounts jointly for the sampling uncertainty of both endpoints. The construction handles only *excess* selection in the treated cell (Section 8).

5.6 Diagnostics, falsification, and the fathers’ comparison

We estimate event-study versions of equation (1) with Mother $\times$ year interactions (reference year 2019) for both outcomes and test the pre-period coefficients jointly. Because those test the difference-in-differences assumption rather than the triple-differences one, we also estimate the event-study version of equation (2) itself — Post replaced by survey-year interactions at every level, so that Mother $\times$ year $\times$ WFH_Exposure is recovered separately for each year — and apply the same joint test to its pre-period coefficients.

The falsification test is a permutation test on the exposure score. Under the sharp null that teleworkability is unrelated to the post-2021 change in the motherhood gap, the forty occupation scores are exchangeable across occupations. We therefore reassign the forty scores at random across the forty occupation codes 999 times, holding every woman’s occupation and the mother/period structure fixed, refit equation (2) each time, and record the cluster-robust t -statistic of the triple interaction. The permutation p -value is the share of draws, counting the observed one, whose $|t|$ reaches the observed value (Phipson and Smyth, 2010). Because it is built from the design itself, it needs neither the $t(39)$ approximation nor a bootstrap, and it answers the question a placebo should: how often would a design like this one produce an exposure gradient as large as ours when exposure is, by construction, noise.

The remaining checks re-estimate the hours DDD with an added Mother $\times$ age-group interaction and with pre-period age-group raking weights, in response to the age imbalance noted above; with the external and realized exposure indices in place of the calibrated one; with alternative codings of the hours outcome; separately for Jewish and Arab women; by the age of the youngest child; and on the male subsample with “mother” read as “father.” The last of these is a comparison, not a placebo: fathers in teleworkable occupations are exposed to remote work exactly as mothers are, so a null for fathers is not what the design predicts, and a negative gradient would be evidence on household reallocation rather than on a confound. Coefficients across subsamples are compared with two-sample z -tests, which are valid without a covariance correction because the samples are disjoint.

6 Results

Throughout, significance is denoted *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, · $p < 0.1$. Standard errors are in parentheses.

6.1 Pre-trends and the hours DiD

Before turning to the estimates, Figure 3 reports the hours event study. Relative to 2019, neither pre-period Mother $\times$ year coefficient is distinguishable from zero — 0.3515 in 2017 and 0.2013 in 2018 — and a joint Wald test on the two does not reject ($F(2, 63, 198) = 0.949$, $p = 0.387$). The same test on the employment outcome also passes (Section 6.8), so the difference-in-differences identifying assumption is supported on both margins over the full pre-period. The triple difference rests on a different and stricter assumption, which requires an event study of its own (Section 6.3). In the post-period the effect builds gradually — 0.3923 in 2021, 0.1213 in 2022, and 0.7517** in 2023 — and Figure 1 shows the corresponding raw gaps with no regression adjustment at all.

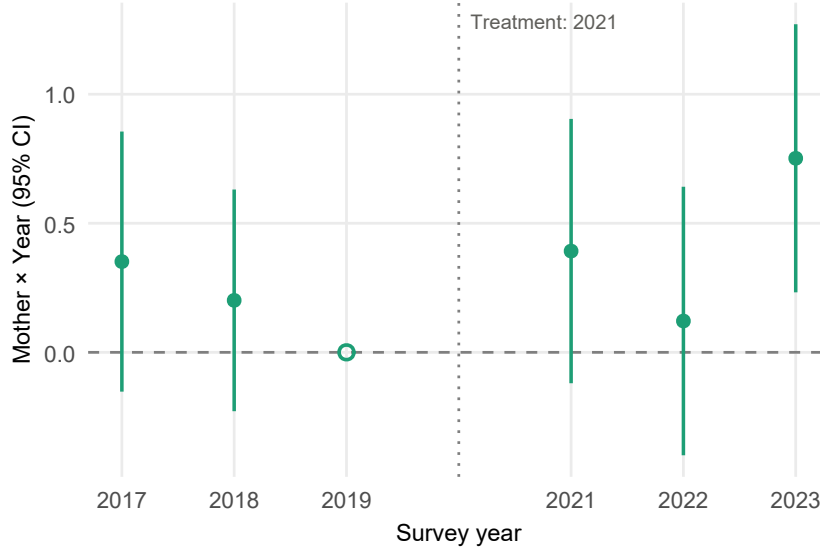


Figure 3: Event study: Mother $\times$ year interactions for weekly hours, reference year 2019

Notes: Women working in the reference week ($N = 251,857$); standard controls; standard errors clustered by individual. Vertical bars are 95% confidence intervals. Coefficients (SE): 2017, 0.3515 (0.2570); 2018, 0.2013 (0.2191); 2021, 0.3923 (0.2612); 2022, 0.1213 (0.2654); 2023, 0.7517** (0.2650). Joint test on 2017–2018: $F(2, 63, 198) = 0.949$, $p = 0.387$. The hollow point at 2019 is the omitted reference year, pinned at zero by construction rather than estimated. The gap between 2019 and 2021 is the excluded 2020 survey year; the dotted rule marks the treatment boundary.

Column (1) of Table 2 reports equation (1) for weekly hours among women working in the reference week. Mothers work 1.739 fewer hours per week than childless women

(Mother = -1.739^{***}), the intensive-margin penalty itself. That penalty did not measurably change after the shift to remote work: Mother $\times$ Post = 0.2280 (SE 0.1809), $t = 1.26$, $N = 251,857$. The raw cell means of Section 4.2 agree on magnitude: the unadjusted difference-in-differences is 0.266, so the controls move the estimate by less than four hundredths of an hour. They do not agree on precision — the unclustered standard error is 0.098, roughly half the clustered one — which is why the raw arithmetic should not be read as evidence of a significant increase. At this standard error the minimum detectable effect is $2.8016 \times 0.1809 \approx 0.51$ hours, so the estimate rules out average gains much above half an hour a week: a reasonably precise null, not an uninformative one. Replacing the single Post dummy with survey-year effects, so that the year-to-year movements of Figure 1 are absorbed rather than pooled, leaves it at 0.2310 (SE 0.1808).

Selection into the estimation sample does not change that reading. The probability that hours are observed for post-period mothers ($s_{11} = 0.6964$) exceeds the parallel-trends-in-selection counterfactual ($s_{11}^* = 0.6786 + (0.7058 - 0.7010) = 0.6833$), so there is excess selection to trim — about 1.9% of the cell.¹ The resulting Lee bounds are -0.5166 (SE 0.1781) and 0.8017 (SE 0.1789), with an Imbens–Manski 95% confidence interval for the identified set of $[-0.810, 1.096]$. The correction widens the interval around a null rather than overturning a finding. We therefore do not treat the plain hours DiD as evidence of an average effect in either direction, and nothing that follows rests on it.

6.2 The hours DDD: the paper’s headline result

Column (2) of Table 2 interacts the DiD with the calibrated occupation-level exposure index. The triple interaction is 3.224^{**} (SE 1.022, $N = 246,326$): among mothers who are working, those in more WFH-exposed occupations saw a larger post-2021 increase in weekly hours relative to childless women in the same occupations — about 3.2 additional hours per unit of exposure. A unit is not a movement the data contain, so the coefficient needs a scale. On the estimation sample the calibrated index has a standard deviation of 0.199 and an interquartile range of 0.203, so the estimate is 0.64 hours per standard deviation of exposure and 0.65 hours per interquartile range. Between the quartiles of Figure 2, whose mean exposures are 0.06 at the bottom and 0.59 at the top, the regression implies a difference of 1.71 hours a week. The raw quartile difference-in-differences of that figure put the same contrast at 1.87 hours (1.55 against -0.32), within a fifth of an hour of the linear model’s implication; that agreement is the paper’s one check on the linear-in-exposure functional form, and it holds even though the regressor’s dispersion is concentrated in the top quartile.

The column’s second row is as informative as its first. At zero exposure — an occupation that cannot be done from home at all — the implied change for mothers is Mother $\times$

¹Derived as $1 - 0.6833/0.6964$ from the selection rates $s_{00} = 0.7010$, $s_{01} = 0.7058$, $s_{10} = 0.6786$ and $s_{11} = 0.6964$; not directly exported by the pipeline.

Post = -0.5385 (SE 0.3364), if anything negative. And because Post $\times$ WFH_Exposure is in the model, the gradient is net of whatever happened to hours in teleworkable occupations for all women. Set against the near-zero average effect of column (1), the reading is not that a broad gain is *larger* in teleworkable jobs. It is that outside teleworkable jobs there is no gain to speak of, and the entire effect is the exposure gradient. That is a sharper claim than the one a significant average DiD would have supported, and it is the claim Figure 2 shows directly: three quartiles at zero and one well above it.

The estimate is 3.15 analytic standard errors from zero; Section 6.4 reports the bootstrap and permutation p -values that qualify that figure. The design’s minimum detectable effect is 2.864 hours per unit of exposure, or 0.57 hours per standard deviation (7.3% of mean weekly hours), so an effect of the size observed is detectable, though without a great deal of room to spare; the reweighted specification of Section 6.4 shows what that margin means in practice.

Columns (3) and (4) relax the functional form. With occupation-by-year, occupation-by-mother and mother-by-year fixed effects in place of every lower-order term (equation (3)), the triple interaction is 2.876 (SE 0.9495), about a tenth below the pooled estimate and still significant at the 1% level on the analytic standard error: the linear lower-order terms of column (2) are not what produces the result. Column (4) replaces the score with indicators for the pre-period exposure quartile the occupation falls in, the bins of Figure 2. Relative to the least teleworkable quartile, the post-2021 change in the mother/non-mother gap is 0.75 hours (SE 0.48) in the second quartile, 0.60 (SE 0.48) in the third, and 1.844 (SE 0.594, $p < 0.01$) in the top quartile. The top-quartile contrast is the between-quartile difference the linear model implies (1.71) and the raw figure shows (1.87), now carrying a standard error, and it is the only quartile contrast that is individually significant: the response is a step at the top of the exposure distribution more than a smooth gradient across it. That top quartile holds six of the forty occupations (sixteen, seven, eleven and six from the bottom up), which is why it matters that the pooled and saturated columns, which use all forty, agree with it in sign and magnitude.

6.3 Diagnostics for the triple difference

Pre-trends. The pre-trend test of Figure 3 licenses equation (1), not equation (2). It asks whether mothers and childless women trended together; the triple difference requires the stricter condition that the mother/non-mother gap trended together *across levels of WFH exposure*. These are not the same assumption, and the difference-in-differences version is not the conservative one: because it averages over exposure, a pre-period divergence correlated with exposure but netting to zero across occupations would leave the DiD test untouched while invalidating the headline estimate. Figure 4 therefore reports the event-study analogue of equation (2), on the same sample, exposure index and clustering

Table 2: Intensive-margin DiD and DDD: weekly hours, reference-week workers

	(1) DiD	(2) DDD	(3) DDD, saturated	(4) DDD, quartiles
Mother $\times$ Post $\times$ WFH_Exposure		3.224** (1.022)	2.876** (0.9495)	
Mother $\times$ Post $\times$ Q2 (vs. Q1)				0.7529 (0.4827)
Mother $\times$ Post $\times$ Q3 (vs. Q1)				0.5998 (0.4847)
Mother $\times$ Post $\times$ Q4 (vs. Q1)				1.844** (0.5946)
Mother $\times$ Post	0.2280 (0.1809)	−0.5385 (0.3364)		−0.4720 (0.3459)
Mother $\times$ WFH_Exposure		−1.295 (1.053)		
Post $\times$ WFH_Exposure		−0.5483 (0.8102)		
WFH_Exposure		1.128 (3.912)		
Mother	−1.739*** (0.1441)	−1.449*** (0.3512)		−1.527** (0.4523)
Post	−0.0570 (0.1481)	0.0535 (0.2732)		−0.1305 (0.3666)
Constant	32.35*** (0.3182)	32.12*** (1.540)		31.58*** (2.051)
MDE for triple interaction		2.8640		
SD of exposure, estimation sample		0.1986		
Triple interaction per SD of exposure		0.640		
Occupations in Q1 / Q2 / Q3 / Q4				16 / 7 / 11 / 6
Fixed effects	None	None	Three sets	None
Observations	251,857	246,326	246,324	246,326
R^2	0.04049	0.04094	0.13512	0.04815
Clustering	individual	occupation (40)	occupation (40)	occupation (40)

Dependent variable: usual weekly work hours, women working in the reference week. Standard controls in every column. Columns (2)–(4) use the calibrated occupation-level exposure index on column (1)’s sample less the rows with no matched 2-digit occupation. Column (3) is equation (3): the “three sets” of fixed effects are occupation $\times$ year, occupation $\times$ mother and mother $\times$ year, which absorb every lower-order term, so only the triple interaction is reported. Column (4) replaces the score with indicators for its pre-period quartile (Figure 2’s bins) and reports the three quartile contrasts against the bottom quartile; its lower-order quartile terms are estimated but not shown, and the number of occupations in each bin is listed because occupation is the cluster. MDE at $\alpha = 0.05$, power 0.80. Columns (2)–(4) use a t reference distribution with 39 degrees of freedom (Section 5.4); the wild-bootstrap and permutation p -values for column (2)’s triple interaction are in Table 4. Column (2)’s Mother $\times$ Post is the effect at *zero* exposure, not an average effect. No coefficient is dropped for collinearity in any column.

as column (2) of Table 2.

The result is supportive. Neither pre-period coefficient is close to significant, and neither is far from zero in level: 0.5866 in 2017 and 0.3134 in 2018, against post-period estimates five to thirteen times larger. The joint Wald test does not reject, and does not come near rejecting ($F(2, 39) = 0.041$, $p = 0.960$). The post-period path is flat, then steps up in 2021 and rises: 3.011*, 3.545**, 4.020**, all three individually significant and straddling the pooled 3.224 of Table 2, which is what a treatment beginning in 2021 looks like rather than a pre-existing trend carried through the sample.

One caveat belongs with this test rather than in a footnote. Clustering at the level the exposure regressor actually varies at leaves forty occupation clusters, so the pre-period standard errors (2.2367 and 1.7449) are of the same order as the effect being tested. Each pre-period confidence interval therefore contains values larger in magnitude than 3.224: the test is consistent with parallel trends and shows no sign of a violation, but it does not have the power to rule out a pre-trend as large as the estimate itself. The wild cluster bootstrap does not change that reading: the bootstrap p -values are 0.84 for the 2017 coefficient, 0.88 for 2018 and 0.85 for their sum. This is the same small-cluster limitation the headline estimate carries, and we read it as supporting evidence rather than as a closed question.

Lee bounds. Table 3 applies the generalized selection correction. Excess selection into the estimation sample among post-period mothers is present in all four quartiles of the cell-based exposure index but modest everywhere: the trim proportion ranges from 0.80% (quartile 3) to 4.48% (quartile 1, the least teleworkable jobs), and a total of 1,396 of 72,820 post-period working mothers are trimmed. The bounds move little: the lower bound is 3.1255 (SE 1.1440), the upper bound 3.4434 (SE 1.0220), and the Imbens–Manski 95% confidence interval for the identified set is [1.021, 5.324], which excludes zero. The triple interaction therefore survives the selection correction, and is the strongest single piece of evidence in the paper.

6.4 Robustness

Table 4 collects the remaining checks.

Exposure measure. With the raw Dingel and Neiman (2020) external index in place of the calibrated one, the triple interaction is 0.6720 (SE 0.8838), positive but insignificant; with the realized 2021-anchored index it is 5.195* (SE 2.363). The direction is consistent across all three measures but the magnitude and significance are not, and we describe the result as directionally rather than point-estimate robust to the exposure measure.

That pattern deserves a direct statement, because it is the natural objection to the design. The external index is the only one of the three built entirely from pre-treatment

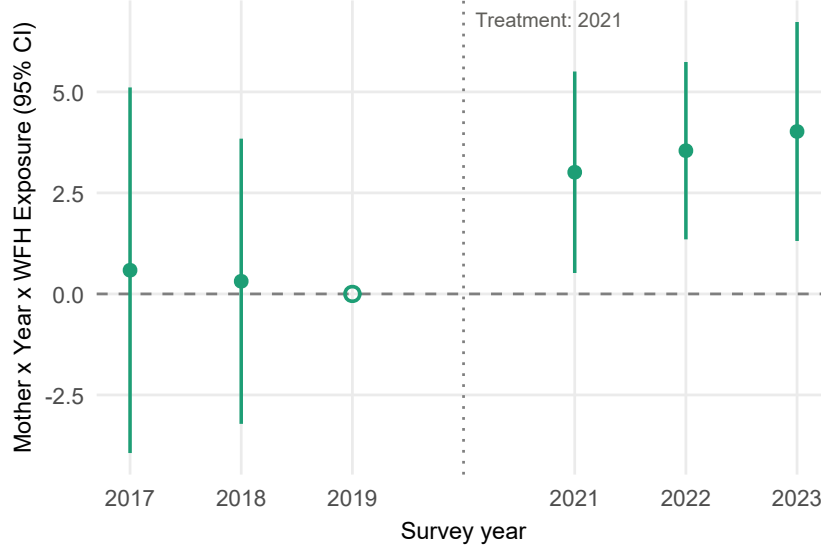


Figure 4: Event study for the triple difference: Mother $\times$ year $\times$ WFH_Exposure interactions for weekly hours, reference year 2019

Notes: Employed women matched to a two-digit occupation and working in the reference week ($N = 246,326$); standard controls; calibrated occupation-level exposure; standard errors clustered by two-digit occupation (40 clusters). Vertical bars are 95% confidence intervals. Coefficients (SE), hours per unit of exposure: 2017, 0.5866 (2.2367); 2018, 0.3134 (1.7449); 2021, 3.0114* (1.2327); 2022, 3.5454** (1.0850); 2023, 4.0203** (1.3397). Joint test on 2017–2018: $F(2, 39) = 0.041$, $p = 0.960$; wild cluster bootstrap p -values (Rademacher, 9,999 draws) 0.838 for 2017, 0.884 for 2018 and 0.852 for their sum. The specification includes the year main effects and both two-way year interactions (Mother $\times$ year and WFH_Exposure $\times$ year) alongside the triple interaction plotted here; omitting any of them would make the estimate a two-way difference rather than a triple one. The hollow point at 2019 is the omitted reference year. The gap between 2019 and 2021 is the excluded 2020 survey year; the dotted rule marks the treatment boundary. Compare Figure 3, which plots the difference-in-differences analogue and tests a weaker assumption.

Panel A: selection rates and trimming by cell-exposure quartile

Table 3: Generalized Lee bounds on the hours DDD (Mother $\times$ Post $\times$ WFH_Exposure)

Quartile	s_{00}	s_{01}	s_{10}	s_{11}	s_{11}^*	$n(\text{Mother}=1, \text{Post}=1)$	Trim
1	0.5830	0.5862	0.5497	0.5788	0.5529	21,655	4.48%
2	0.7070	0.7120	0.6856	0.6977	0.6906	28,540	1.01%
3	0.7542	0.7450	0.7554	0.7522	0.7462	34,293	0.80%
4	0.8194	0.8029	0.7352	0.7399	0.7188	21,815	2.85%

Panel B: bounds on the triple interaction

Bound	Coefficient	SE	95% CI
Lower	3.1255	1.1440	[0.8833, 5.3677]
Point (untrimmed)	3.2319	1.0538	[1.1664, 5.2973]
Upper	3.4434	1.0220	[1.4404, 5.4465]
Imbens–Manski 95% CI for the identified set			[1.021, 5.324]

s_{ab} is the probability that hours are observed — employed *and* working in the reference week — so these rates are lower than the corresponding employment rates. Panel A’s n column counts all post-period mothers in the quartile, working or not, because the cell-based index is defined for the non-employed as well; the trim proportions apply to the occupation-matched subset with observed hours (1,396 of 72,820 in total), which is the population the bound regressions use. The untrimmed point estimate here (3.2319) differs slightly from Table 2 (3.224) because this sample additionally requires a matched cell-based exposure. Outcome regressions use the occupation-level index and cluster by occupation.

information, and it yields the weakest estimate; the two that draw on post-period Israeli data yield significant ones. One reading, which we hold, is that the mechanism becomes visible only once exposure is measured as Israeli jobs were actually done rather than as US task content says they could be: teaching and clerical support, which the external index scores as almost fully teleworkable, returned fully to the workplace, and an index that places them in the treated tail is measuring the wrong thing. The other reading is that calibrating on 2022–23 realized WFH lets the regressor absorb post-period outcomes. Four features of the construction limit that risk: the realized shares are computed on men and women together, not on the women in the regression; only ten of forty occupations are swapped, and only where the gap exceeds 0.5 and a one-sided test at the 95% level rejects sampling noise; all ten swaps are *downward*, removing occupations from the treated tail rather than adding any to it; and the outcome is usual hours, not the WFH indicator the calibration uses. The direct test is to drop the ten swapped occupations and re-estimate on the thirty where the calibrated and external indices coincide by construction, so that no post-period information enters the regressor at all. On that sample the triple interaction is 4.309** (SE 1.192, $N = 132,046$, 30 clusters), larger than the headline, not smaller. The post-period calibration does not produce the result; if anything the ten swapped occupations, which together account for nearly half the estimation sample, dilute it. What the calibration cannot be made into is a pre-treatment measure.

Age balance. Adding a Mother $\times$ age-group interaction leaves the estimate essentially unchanged (3.205**, SE 1.055). Reweighting the sample to balance pre-period age groups between mothers and non-mothers attenuates it by about 19% to 2.597* (SE 1.050) — still significant, though at the 5% level rather than the 1%, and a value the design would detect only narrowly (its minimum detectable effect is 2.864), so we would not rest the result on this specification alone. The imbalance being corrected is real and tracks exposure: the pre-period mother/non-mother gap in the age-group code is -0.974 in the lowest exposure quartile ($t = -94.0$), -0.735 in the second, -0.169 in the third, and $+0.057$ in the highest. The reweighted attenuation is the largest single move any check produces, and it is worth reporting for that reason.

Sample. Excluding the 2023 survey year, whose final quarter includes the start of the October 2023 war (Section 8), the triple interaction is 2.979* (SE 1.170, $N = 208,833$): smaller, since 2023 carries the largest event-study coefficient, but still significant at the 5% level, while the plain hours DiD stays a null (0.0710, SE 0.1975).

Outcome coding. The hours variable is a bin-median approximation with two irregular-hours codes imputed from period-specific medians (Section 3), and the effect is smaller than one bin width, so the result should not depend on how the bins are turned into

hours. Dropping the two imputed codes instead of imputing them gives 3.056** (SE 1.059, $N = 232,785$). Because the bin medians make 35 and 40 hours exact bin edges, the same specification can be estimated as a linear probability model on bin-crossing outcomes: the probability of usually working at least 35 hours rises by 0.0964** (SE 0.0291) per unit of exposure for mothers relative to childless women after 2021, and the probability of working at least 40 hours by 0.1073** (SE 0.0380). Per standard deviation of exposure those are 1.9 and 2.1 percentage points, against pre-period shares among working mothers of 71% working at least 35 hours and 61% at least 40. The effect is a shift of mothers in teleworkable jobs across the full-time boundary, which is the reading Section 3 anticipated.

Inference. Clustering two ways, by individual and by occupation, leaves the headline standard error unchanged at 1.022: the occupation-level dependence already dominates the individual-level one. The forty-cluster problem is not addressed by two-way clustering, and the two procedures of Section 5.4 that do address it disagree with the analytic p -value of 0.003. The wild cluster bootstrap gives $p = 0.056$, with a bootstrap 95% confidence set of $[-0.26, 5.84]$; the permutation test gives $p = 0.032$, with 31 of 999 random reassignments of the forty exposure scores producing a $|t|$ at least as large as the observed 3.15 (Section 6.5). The estimate is therefore significant at the 1% level on the analytic standard error, at the 5% level under the permutation test, and at the 10% level under the bootstrap, and we report all three rather than choosing among them. The bootstrap, which re-signs whole-cluster residuals, is the most conservative of the three; the permutation test holds the design’s cluster structure fixed and randomizes only the regressor, which is the sharper question for a triple difference whose identifying variation is occupation-level exposure.

6.5 Falsification: permuting exposure across occupations

Figure 5 plots the reference distribution the permutation test builds. Each of the 999 draws reassigns the forty calibrated scores at random across the forty occupation codes, leaving every woman’s occupation, her mother status and the survey year untouched, and refits equation (2) with occupation-clustered standard errors. Under this sharp null the design produces cluster-robust t -statistics whose central 95% lie between -2.96 and 2.60 , with the largest $|t|$ of any draw at 6.21; the observed 3.15 is exceeded in absolute value by 31 draws, so $p = 0.032$ (Monte-Carlo standard error 0.006). Read the other way, a design with this cluster structure produces an exposure gradient as large as ours about three times in a hundred when exposure carries no information. That is the falsification a placebo should supply and a fathers’ regression cannot: the latter compares a different, also-treated population, whereas this test asks the design itself how often it finds a gradient

Table 4: Hours DDD: sensitivity of Mother $\times$ Post $\times$ WFH_Exposure

Specification	Coefficient	SE	p	N
<i>Exposure measure</i>				
Calibrated (primary)	3.224**	1.022	0.0031	246,326
External (Dingel–Neiman)	0.6720	0.8838	0.4517	246,326
Realized (2021 anchor)	5.195*	2.363	0.0346	246,037
<i>Age balance</i>				
Age-interacted (+ Mother $\times$ age group)	3.205**	1.055	0.0042	246,326
Reweighted (pre-period age-group raking)	2.597*	1.050	0.0178	242,552
<i>Sample</i>				
Unswapped occupations only (30 of 40)	4.309**	1.192	0.0011	132,046
Excluding survey year 2023	2.979*	1.170	0.0151	208,833
<i>Outcome coding</i>				
Irregular-hours codes dropped, not imputed	3.056**	1.059	0.0064	232,785
Full-time indicator (≥ 35 hours)	0.0964**	0.0291	0.0020	246,326
Long-hours indicator (≥ 40 hours)	0.1073**	0.0380	0.0075	246,326
<i>Inference on the primary estimate</i>				
Two-way clustering (individual, occupation)	3.224**	1.022	0.0031	246,326
Wild cluster bootstrap ($B = 9,999$)	3.224	1.022	0.0558	246,326
Permutation test (999 reassignments)	3.224		0.0320	246,326

All rows: the triple interaction Mother $\times$ Post $\times$ WFH_Exposure from equation (2), women working in the reference week, standard controls, calibrated exposure unless stated, clustered by occupation unless stated; p is the analytic $t(39)$ p -value unless the row says otherwise. The unswapped-occupations row drops the ten occupations whose exposure was swapped to its realized 2022–23 value, so the calibrated and external indices coincide on the rows that remain. The two indicator outcomes are linear probability models on bin-crossing outcomes: the bin medians make 35 and 40 hours exact bin edges. The no-imputation row drops the two irregular-hours codes instead of imputing them. The wild-bootstrap row reports the analytic coefficient and standard error with the bootstrap p -value (Section 5.4); the permutation row reports the permutation p -value of Section 5.6, which has no standard error.

in noise.

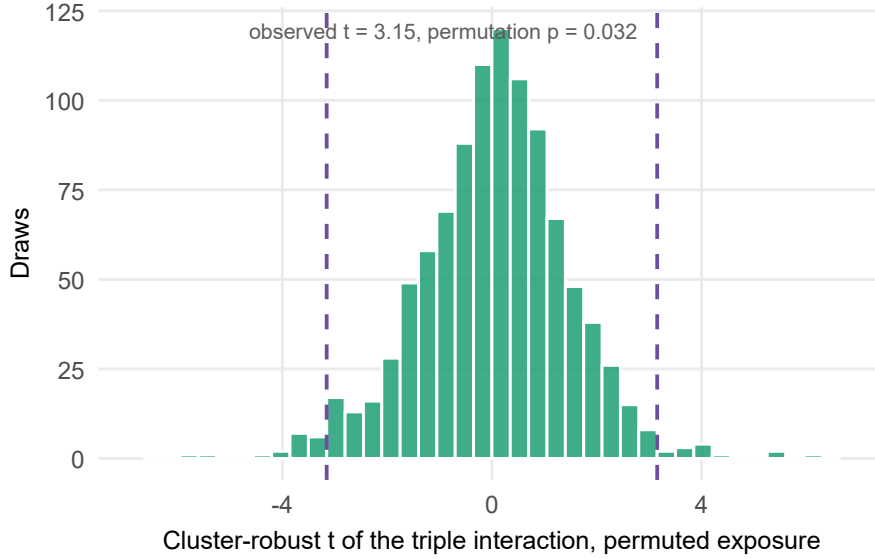


Figure 5: Permutation distribution of the hours DDD triple interaction
Notes: Histogram of the cluster-robust t -statistic of Mother $\times$ Post $\times$ WFH_Exposure over 999 random reassignments of the forty occupation-level exposure scores across occupations (seed fixed in the replication code), with the same sample, controls and clustering as column (2) of Table 2. Dashed lines mark $\pm$ the observed statistic (3.15); $p = 0.032$ counts the observed draw (Phipson and Smyth, 2010).

6.6 Heterogeneity by the age of the youngest child

If remote work relaxes a time constraint that binds through child care, the response should be largest for mothers of young children and smallest for mothers whose youngest child is a teenager. The survey records the youngest child’s age in bins, so Table 5 and Figure 6 re-estimate the hours DiD and DDD four times, each time comparing one group of mothers — those whose youngest child is aged 0–4, 5–9, 10–14 or 15–17 — with the same 133,646 childless women. The average effect is a null in every group: the DiD is between 0.13 and 0.56 hours and never distinguishable from zero. The exposure gradient declines monotonically with the child’s age: 3.656** hours per unit of exposure (SE 1.079) for mothers of under-fives, 3.211* (SE 1.311) for youngest children aged 5–9, 2.741 (SE 1.525, $p = 0.08$) for 10–14, and 1.862 (SE 0.935, $p = 0.05$) for 15–17, half the under-five estimate. Because the four regressions share the control group they are not independent samples and the two-sample z -test of Section 5.6 does not apply across rows; the individual confidence intervals overlap, so the pattern is a gradient consistent with the mechanism rather than a set of statistically separated effects. It also answers the concern that “mother” is a broad category: the estimate is not carried by mothers of teenagers, for whom the time constraint is weakest, and it is largest exactly where the mechanism

predicts.

Table 5: Hours DiD and DDD by age of the youngest child

Sample of mothers	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure		
	Coef.	SE	N	Coef.	SE	N
All mothers (Table 2)	0.2280	0.1809	251,857	3.224**	1.022	246,326
Youngest child aged 0–4	0.1267	0.2111	158,424	3.656**	1.079	154,499
Youngest child aged 5–9	0.1621	0.2533	130,782	3.211*	1.311	127,419
Youngest child aged 10–14	0.1593	0.2780	123,280	2.741	1.525	120,084
Youngest child aged 15–17	0.5613	0.3483	107,061	1.862	0.9354	104,079

Weekly hours, reference-week workers, standard controls; each row compares the named group of mothers (youngest child’s age from the CBS bin variable) with all childless women, so the control group is identical across rows and the rows are not independent samples. DiD clustered by individual; DDD clustered by occupation, calibrated exposure. The first row repeats Table 2. Mothers whose youngest-child code falls outside the four bins enter no row. No coefficient is dropped for collinearity in any column.

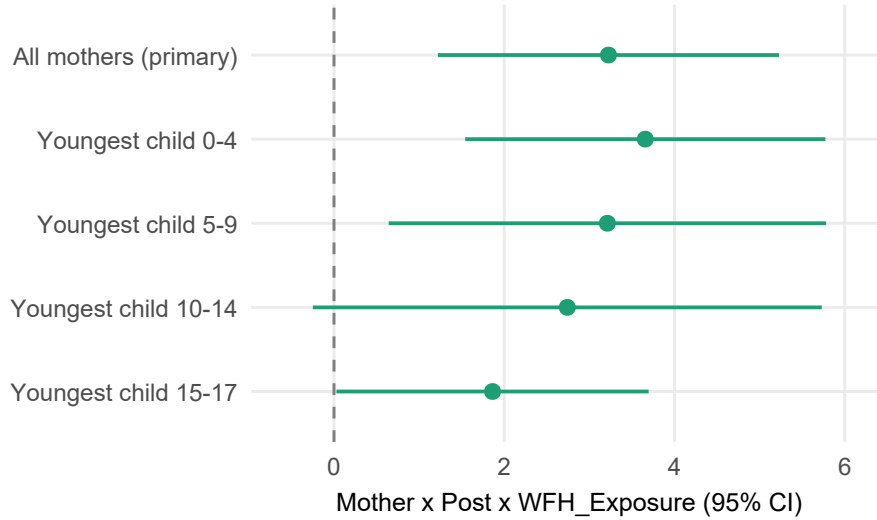


Figure 6: Hours DDD triple interaction by age of the youngest child
Notes: The DDD column of Table 5: Mother $\times$ Post $\times$ WFH_Exposure with 95% confidence intervals clustered by occupation, one estimate per youngest-child age bin, each against all childless women. The top row is the pooled estimate of Table 2.

6.7 Heterogeneity by population group, and the fathers’ comparison

Table 6 re-estimates the hours DiD and DDD separately for Jewish and Arab women, and on men as a comparison population. For Jewish women the pattern is the pooled pattern: the DiD is 0.3021 (SE 0.2027), indistinguishable from zero, while the DDD is 2.8465** (SE

0.9125). For Arab women the DiD is -0.0724 (SE 0.4785), also indistinguishable from zero — and Arab women’s hours fell in the post-period overall (Post = -2.162^{***}) — while the DDD is 5.5069^* (SE 2.6741), nominally significant and nearly twice the Jewish point estimate. Neither group shows an average effect; both show an exposure gradient.

We do not read the larger Arab estimate as evidence that the mechanism is stronger for Arab women. The two coefficients are not statistically distinguishable: the Arab subsample is about nine times smaller (24,307 versus 209,727 occupation-matched women with observed hours), its standard error nearly three times larger, and a two-sample z -test on the DDD term gives $z = 0.942$, $p = 0.346$. There is also a mechanical reason for the imprecision: Arab women’s employment in Israel is concentrated in a narrow set of occupations, disproportionately education and health-aide roles, so comparatively few two-digit occupations carry enough Arab respondents to inform an occupation-specific comparison, and a small number of high-leverage occupations can dominate the estimate. The Arab result is therefore best described as imprecisely estimated and statistically indistinguishable from the Jewish estimate — a thin-coverage fragility rather than a documented heterogeneous treatment effect.

The fathers’ comparison is a comparison, not a placebo, and has to be read on both margins. Re-estimating the same models on men, with “mother” read as “father,” gives a DiD of -0.4002^* (SE 0.1883, $N = 261,543$) and a DDD of -1.7620 (SE 1.0153, $N = 251,779$), the latter negative and significant at the 10% level. Fathers in teleworkable occupations are exposed to remote work exactly as mothers are, so a null was never the prediction; what the comparison shows is that the exposure gradient has the opposite sign for fathers, and that the two gradients are statistically distinguishable ($z = 3.461$, $p = 0.0005$). A macro trend common to all workers in teleworkable jobs cannot produce a positive gradient for mothers alongside a negative one for fathers, which rules out the most obvious confound; and a negative gradient for fathers is what within-household reallocation would look like if mothers’ additional paid hours in teleworkable jobs were matched by fathers’ additional care hours. We report that as a substantive reading the data are consistent with, not as a result, since the design is not built to identify it. The falsification test proper is Section 6.5.

On the plain DiD fathers’ hours fall by 0.40 relative to childless men, a larger and more significant movement than the 0.23 we estimate for mothers, and although the two are distinguishable ($z = 2.406$, $p = 0.016$) the difference is driven by the fathers’ coefficient. We do not claim an average effect for mothers on that margin; Section 6.1 reads the female DiD as a null, and the fathers’ comparison is consistent with that reading.

Table 6: Hours DiD and DDD by subgroup, and the fathers’ comparison

Subgroup	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure		
	Coef.	SE	N	Coef.	SE	N
All women (primary)	0.2280	0.1809	251,857	3.224**	1.022	246,326
Jewish women	0.3021	0.2027	214,449	2.847**	0.9125	209,727
Arab women	-0.0724	0.4785	24,975	5.507*	2.674	24,307
Men (fathers vs. childless)	-0.4002*	0.1883	261,543	-1.762	1.015	251,779
95% CI, Jewish	[-0.0952, 0.6994]			[1.0579, 4.6351]		
95% CI, Arab	[-1.0103, 0.8656]			[0.2657, 10.7481]		
Jewish vs. Arab, z (p)	0.720 (0.471)			-0.942 (0.346)		
Women vs. men, z (p)	2.406 (0.016)			3.461 (0.001)		

Weekly hours, reference-week workers, standard controls. DiD clustered by individual; DDD clustered by occupation. Men: “Mother” read as “Father,” a comparison population rather than a placebo (Section 5.6). Two-sample z -tests as described there. Confidence intervals use the normal critical value. The Arab-women DiD drops the religion category ‘other’, constant on that sample; the Arab-women DDD drops the religion category ‘other’, constant on that sample; no other column drops a coefficient.

6.8 The extensive margin, briefly

The employment margin is where this sample has the least room to move — mothers’ employment rate exceeds childless women’s in the raw data — and it is secondary for a reason of power that is worth stating once. Column (1) of Table 7 reports equation (1) with employment as the outcome: Mother $\times$ Post = -0.0053 (SE 0.0052). The mother/non-mother employment gap did not change after the shift to remote work, and the estimate is precise — its standard error is about half a percentage point — so this is an informative null. Employment rose in the post-period for all women (Post = 0.0153^{***}); the DiD estimated separately for Jewish and Arab women is -0.0041 (SE 0.0058) and -0.0208 (SE 0.0138), neither significant; and the pre-period event-study coefficients pass their joint test ($F(2, 79,069) = 0.63$, $p = 0.53$).

Column (2) reports the employment DDD with the cell-based exposure index. The triple interaction is -0.0257 (SE 0.0725); a variant with fully interacted age-by-education-by-district cell fixed effects gives -0.0194 (SE 0.0722). This null is of a different kind. At $\alpha = 0.05$ and 80% power the design could detect a triple interaction no smaller than 0.2031 per unit of exposure — and the cell-based index never moves a unit: its standard deviation on the estimation sample is 0.0706 and its interquartile range 0.0766, so the detectable effect is 1.43 percentage points per standard deviation of exposure, or 1.85% of the baseline employment rate of 0.7737. Against the closest published benchmark, [Harrington and Kahn](#)’s 0.78 percentage points per 10% rise in WFH, the design is underpowered by a factor of about two to three. The shortfall is structural — a binary outcome and a regressor with a standard deviation of 0.07 — and re-clustering, combining the two margins into a single outcome, and refining the exposure-cell partition did not close it. The employment DDD is therefore *uninformative*: it is evidence neither for nor against a WFH-mediated

employment effect, and the two nulls in Table 7 should not be read as one verdict. The one significant coefficient, $\text{Post} \times \text{WFH_Exposure} = -0.1555^{**}$, indicates a general post-2021 employment decline in high-exposure cells that is unrelated to motherhood. This is what motivates the intensive-margin design, where occupation-level exposure is available.

Table 7: Extensive margin: employment DiD and DDD

	(1) DiD	(2) DDD, cell-based exposure
Mother $\times$ Post $\times$ WFH_Exposure		-0.0257 (0.0725)
Mother $\times$ Post	-0.0053 (0.0052)	0.0065 (0.0168)
Mother $\times$ WFH_Exposure		-0.1404 (0.0803)
Post $\times$ WFH_Exposure		-0.1555** (0.0566)
WFH_Exposure		0.0131 (0.0565)
Mother	-0.0034 (0.0041)	
Post	0.0153*** (0.0043)	
Constant	0.4926*** (0.0085)	
MDE ($\alpha = 0.05$, power 0.80), per unit		0.2031
MDE per SD of exposure		0.0143
as % of baseline employment rate		1.85%
Observations	364,784	355,984
R^2	0.22689	0.22103
Clustering	individual	demographic cell (≈ 210)

Dependent variable: employment indicator; full sample of employed and non-employed women. Standard controls in both columns; column (2) adds Mother $\times$ age-group interactions, so its Mother and Post main effects are not separately reported. Column (2) uses the demographic-cell shift-share index of Section 3.2, whose standard deviation on this sample is 0.0706 and interquartile range 0.0766; the per-unit MDE therefore answers a question the data cannot pose, and the per-SD row is the interpretable one. Baseline employment rate 0.7737. No coefficient is dropped for collinearity in any column.

7 Discussion

Among mothers who remain in work, there is no post-2021 hours effect on average. There is a large one in teleworkable occupations. Those two statements are the paper’s result, and it is the conjunction that makes it informative: a triple-interaction coefficient of about 3.2 additional weekly hours per unit of exposure, an implied change at zero exposure that is if anything negative, and a selection correction the estimate survives. This is

consistent with WFH access relaxing a binding time constraint specifically for mothers in teleworkable jobs — the [Goldin \(2014\)](#) mechanism — and it is difficult to reconcile with a general post-pandemic recovery common to all women, which would have raised hours across occupations rather than only where the work can be done from home. Fathers show the opposite gradient, and the gradient is steepest for mothers of the youngest children.

How precisely the effect is estimated is a separate question from whether it is there, and the paper keeps the two apart. The exposure score varies across forty occupations, and the three procedures that take that seriously do not agree: the analytic standard error puts the estimate at the 1% level, the permutation test at 5%, the wild cluster bootstrap at 10%. The pattern of evidence — a saturated specification that changes nothing, a top-quartile contrast significant on its own, a monotone child-age gradient, an opposite-signed fathers’ gradient and a selection correction that barely moves the bounds — is what makes the finding credible; no single p -value does.

Which occupations are doing the work is concrete. The calibration step moved ten occupations away from their US task-based scores toward realized Israeli behavior: clerical support and teaching were corrected sharply downward, because Israeli institutional practice kept them in person regardless of task content, while ICT professionals remained high on both measures. The triple interaction is identified off exactly this contrast, which grounds “exposure” in checkable occupational content. That the raw external index yields a positive but insignificant estimate is consistent with that reading, and the check of [Section 6.4](#) that drops the ten calibrated occupations altogether — and finds a larger estimate on the thirty that remain — shows the calibration is not what produces the result.

The magnitude is economically meaningful where it applies. For a mother at the top quartile’s mean exposure the regression implies a post-2021 change of about 1.4 hours a week ($-0.54 + 3.22 \times 0.59$), and the raw top-quartile difference-in-differences of [Figure 2](#) is 1.55; the difference between the top and bottom quartiles is 1.7 hours by the regression and 1.9 by the raw cell means. Set against the 1.74-hour intensive-margin penalty of [Table 2](#), mothers in the most teleworkable jobs recovered roughly four fifths of the penalty after 2021, and mothers in the least teleworkable jobs recovered none of it. The size is also what the most literal form of the time-constraint mechanism would produce: [Aksoy et al. \(2023\)](#) put the commuting time saved at about 72 minutes per day worked from home, so two remote days a week free roughly two and a half hours, and a mother who returns most of that time to paid work is a mother whose hours rise by about the amount estimated here.

The timing is also informative. The event study rises gradually from an insignificant 2021 estimate to a significant one in 2023, consistent with WFH adoption and household time reallocation being a multi-year adjustment rather than an instantaneous response to the 2020–21 shock. The 2023 coefficient is the largest on both margins, and [Section 8](#) notes a reason to hold that year’s estimate a little more loosely than the others.

Why hours and not employment? The employment DDD is not a failed test of the mechanism; it is an uninformative one, underpowered by a factor of two to three against the closest published effect. The intensive-margin design is better powered because it can use occupation-level exposure, which the employment design cannot without conditioning its own outcome on itself. Substantively, the two results are coherent: the effect operates on *how much* mothers in teleworkable jobs work, not on *whether* they are employed, which is precisely the margin [Kleven et al. \(2019b\)](#) identify as a leading channel of the child penalty. This is also where our findings depart from [Harrington and Kahn \(2025\)](#): their US evidence is on employment and income, and we cannot confirm or reject an employment effect in Israel with this design. The policy reading is correspondingly narrow: WFH availability matters for how intensively mothers who already hold *teleworkable* jobs can work, not for whether mothers in general hold jobs or how much they work — which is where our evidence speaks, and where it does not.

The subgroup results are more modest than the point estimates suggest. The Jewish-women estimate carries the pooled result. The Arab-women estimate is larger but imprecise, statistically indistinguishable from the Jewish one, and rests on thin occupational coverage; we report it transparently but do not claim heterogeneity on its basis. The one heterogeneity we do read substantively is by the age of the youngest child, because the mechanism predicts its direction in advance: the gradient is twice as large for mothers of under-fives as for mothers of teenagers, which is what a time constraint that binds through child care should produce.

8 Limitations

Parallel trends. Joint pre-trend tests do not reject on either margin, nor does the triple difference’s own (Figures 3 and 4). The last is the test that matters for the headline estimate and, as Section 6.3 explains, the one with the least power: it is evidence of no visible violation, not proof of none.

The narrowed hours estimand. Hours are usual hours among women who were working, not among all women holding a job, and roughly 10% of each year’s employed sample is excluded on that basis. The exclusion is not innocuous for the triple interaction if absence from work is itself related to WFH exposure, and it is: absence runs at 14.4% of the employed in the least teleworkable quartile of occupations against 8.5–9.6% in the other three, and among mothers it fell after 2021 in the top quartile while rising in the bottom one.² Movement across the absence margin is therefore part of what selection

²Reference-week absence among employed women aged 25–59 with a matched occupation, by the quartiles of Figure 2: 14.4%, 8.5%, 9.5% and 9.6% from the least to the most teleworkable quartile. Among mothers the top-quartile share fell from 12.1% to 10.4% after 2021 and the bottom-quartile share rose from 15.5% to 16.9%; among childless women the top-quartile share is 6.1% in both periods.

into the observed-hours sample means here, which is why the bounds of Section 5.5 define selection on that sample rather than on employment. They trim it; they do not assume it away.

Lee bounds: direction and monotonicity. The trimming construction handles only *excess* selection in the post-period mother cell — the direction the hypothesis predicts — and not under-selection, which is a missing-data problem rather than an excess-observed-data one. It assumes monotone selection to justify trimming from a single tail, and computes selection rates unweighted, consistent with the rest of the analysis. Small excess selection means the construction found little to correct, not that the selection concern is absent.

Survey weights. CBS survey weights are intentionally not applied in any outcome regression. This is a scope decision, not an oversight: reported rates and coefficients are unweighted estimates on the analysis sample, not population-representative statistics. The one exception is internal to constructing the cell-based exposure index, where occupation exposure is aggregated to demographic cells using the survey weight; that is not a representativeness correction for the outcome regressions.

Exposure is partly post-treatment. Neither exposure measure that delivers a significant estimate is fully pre-treatment: the calibrated index draws on 2022–23 realized WFH for its swap test, and the realized index is anchored to 2021 because no 2020 extract exists. Section 6.4 gives the reasons the calibration is unlikely to manufacture the result; the fully pre-treatment external index nonetheless yields a positive and insignificant estimate, and a reader who weights that measure more heavily than we do will read the headline as directionally supported rather than established. The crosswalk behind the external index is itself undocumented (Section 3.2).

The 2023 survey year. The 2023 coefficient is the largest in both event studies, and the 2023 survey year includes the quarter in which the war that began on 7 October 2023 disrupted the labor market: mass reserve call-ups, which bear on men’s hours and so on the fathers’ comparison, and school closures and evacuations, which bear on mothers’. The timing reading of Section 7 does not rest on 2023 alone, since 2021 and 2022 are individually significant on the triple difference, and the estimate survives dropping the year (Section 6.4); the 2023 point should nonetheless be held more loosely than the others. Early 2021 likewise included a third lockdown with partial school closure, so the post-period is “post-acute” more cleanly in its later years than in its first.

Who is childless. The control group is any woman aged 25–59 with no child under 17 in the household, so it includes women whose children are grown and women who

become mothers within the window; [Harrington and Kahn \(2025\)](#)’s design shares this feature. If part of the control group responds to the treatment as mothers do, the DiD is biased toward zero, against the effect we report rather than for it. The treatment group is correspondingly broad, from mothers of infants to mothers of sixteen-year-olds; Section 6.6 splits it by the youngest child’s age.

Age controls and age imbalance. Age enters as a categorical age-group code, because the extract has no continuous age or birth-year variable. Mothers and non-mothers are imbalanced on this code in the pre-period, and the imbalance tracks exposure quartile; the age-interacted and reweighted specifications address it and the estimate survives, but the reweighted attenuation ($3.224 \rightarrow 2.597$) is the largest single move any check produces, so the result is robust to age imbalance in direction and approximate magnitude, not insensitive to it.

Exposure is measured coarsely, and on a mixed scale. Exposure is assigned at the two-digit occupation level, forty groups, and [Buzaglo-Baris \(2023\)](#) shows that within-firm sorting matters as much as occupation for Israeli gender gaps. The calibrated score also mixes task-based probabilities for thirty occupations with realized shares for ten, and Section 3.2 shows that this costs cross-sectional fit: the raw external index predicts realized remote work better in levels. A score on a single scale — realized shares for every occupation — is the realized index of Table 4, which gives a larger and noisier estimate; whether to make it primary is a design choice we flag rather than settle here. Within-occupation heterogeneity in teleworkability is classical measurement error in the regressor, which attenuates the triple interaction toward zero, so the estimate is if anything a lower bound on the effect of actual teleworkability. Occupation codes masked by the CBS for disclosure control are dropped when building exposure; a proxy check finds no strong evidence of bias from this, though that evidence is weak.

Inference. The hours DDD clusters on forty occupation groups, and the three procedures of Section 6.4 disagree on the headline’s significance level: 1% on the analytic standard error, 5% under the permutation test, 10% under the wild cluster bootstrap. That range, not the analytic p -value, is the paper’s statement of how precisely the effect is estimated. The panel does not support within-person identification: only 1.9% of respondents span the pre/post boundary.

9 Conclusion

[SKELETON ONLY — REPLACE WITH YOUR OWN TEXT BEFORE SUBMISSION]

1. **Paragraph 1 — the question and the move.** *Harrington and Kahn (2025) for the US; this paper takes it to Israel and to hours conditional on working. Not a replication (Sections 2, 3).*
2. **Paragraph 2 — the result as a conjunction.** *No average effect: the hours DiD is 0.2280 (SE 0.1809), a reasonably precise null that the selection correction widens rather than overturns. An effect confined to teleworkable work: the triple interaction is 3.224 (SE 1.022, $p < 0.01$), survives the generalized correction (Imbens–Manski [1.021, 5.324]) and two age-balance specifications, and at zero exposure the implied change is negative. Fathers show no exposure gradient, and the women/men difference is $z = 3.461$ (Sections 6.1–6.7).*
3. **Paragraph 3 — what it means.** *Goldin’s time-constraint mechanism: the response is where the work can be done from home, and nowhere else. A result about which mothers benefit, not about mothers as a group. Measurement is part of it: the mechanism is visible only once exposure is measured as Israeli jobs were actually done (Section 7).*
4. **Paragraph 4 — what the design cannot say.** *Employment: the DiD is a precise null, the DDD is uninformative (MDE 1.43 percentage points per SD of exposure, two to three times the Harrington benchmark), and the two should not be conflated. Earnings: not observed. The calibrated exposure is partly post-treatment; the 2023 year includes the war; inference rests on forty clusters; estimates are unweighted (Sections 6.8, 8).*
5. **Paragraph 5 — policy reading and next steps.** *WFH availability bears on how intensively mothers already employed in teleworkable jobs work, not on whether mothers are employed. Next: an earnings outcome, which needs income data linked to the survey since the extract carries none; a higher-powered test of the employment margin; a jointly estimated model letting the mechanism differ by sex and population group.*

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A Appendix: the occupation-level exposure scores

Table A1 lists, for each of the forty two-digit ISCO-08 occupations, the external [Dingel and Neiman \(2020\)](#) score, the realized share usually working from home in 2022–23 that the calibration of Section 3.2 tests it against, whether the score was swapped, the calibrated score the hours DDD uses, and the realized share working from home in the reference week over 2021–23 on men and women pooled. Figure A1 plots the last of these against the

calibrated score. The ten swapped occupations sit above the 45-degree line, because their calibrated value is the usual-place share, which is lower than the reference-week share for every occupation; the thirty unswapped ones lie below it, most of them far below, which is the visual form of the scale-mixing Section 3.2 describes. The five unswapped occupations with calibrated scores above 0.5 (science and engineering professionals; legal, social and cultural professionals and their associate professionals; ICT technicians; numerical and material recording clerks) have realized reference-week shares between 0.15 and 0.38, all below the swapped ICT professionals at 0.62. The correlations and slopes quoted there (calibrated: 0.53 and 0.40 against the reference-week share, 0.48 and 0.26 against the usual-place share; external: 0.74 and 0.37, and 0.59 and 0.21) are computed on these forty points, with the slope weighted by the pooled employed sample in each occupation.

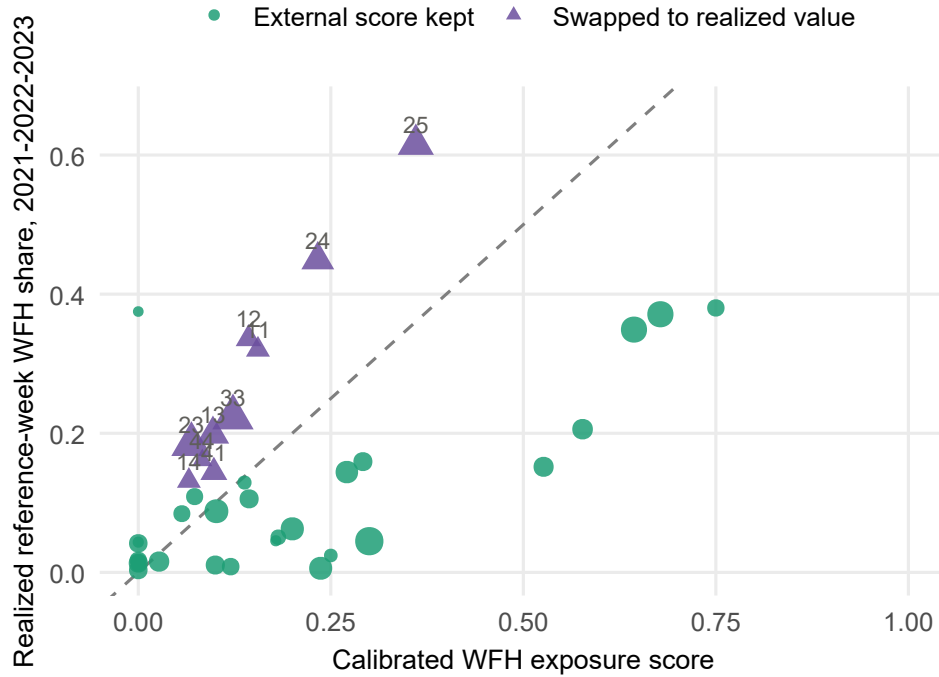


Figure A1: First stage: calibrated exposure score against realized reference-week WFH, by occupation

Notes: One point per two-digit occupation, sized by the pooled employed sample in 2021–23; triangles are the ten occupations the calibration swapped to their realized value, labelled by ISCO-08 code. The dashed line is the 45-degree line. Statistics in the text of Section 3.2.

Table A1: External, realized and calibrated WFH exposure by two-digit occupation

ISCO	Occupation (ISCO-08 sub-major group)	Ext.	Real. 22–23	Swap	Calib.	Ref.-wk 21–23	<i>N</i>
11	Chief executives and legislators	0.900	0.155	yes	0.155	0.320	2,855
12	Administrative and commercial managers	0.892	0.143	yes	0.143	0.336	3,859
13	Production and specialised services managers	0.638	0.097	yes	0.097	0.199	9,990
14	Hospitality, retail and other services managers	0.773	0.066	yes	0.066	0.132	2,827
21	Science and engineering professionals	0.644	0.210		0.644	0.349	13,228
22	Health professionals	0.101	0.059		0.101	0.088	9,665
23	Teaching professionals	0.966	0.069	yes	0.069	0.184	20,496
24	Business and administration professionals	0.943	0.233	yes	0.233	0.449	10,479
25	ICT professionals	1.000	0.360	yes	0.360	0.615	14,638
26	Legal, social and cultural professionals	0.678	0.215		0.678	0.371	13,344
31	Science and engineering associate professionals	0.144	0.059		0.144	0.106	3,926
32	Health associate professionals	0.057	0.080		0.057	0.084	2,203
33	Business and administration associate professionals	0.684	0.123	yes	0.123	0.223	22,415
34	Legal, social and cultural associate professionals	0.577	0.142		0.577	0.206	5,480
35	ICT technicians	0.750	0.244		0.750	0.380	2,491
41	General and keyboard clerks	1.000	0.098	yes	0.098	0.144	4,846
42	Customer services clerks	0.292	0.096		0.292	0.159	3,875
43	Numerical and material recording clerks	0.526	0.071		0.526	0.152	4,948
44	Other clerical support workers	0.625	0.083	yes	0.083	0.161	1,049
51	Personal service workers	0.271	0.146		0.271	0.144	7,808
52	Sales workers	0.200	0.050		0.200	0.063	8,704
53	Personal care workers	0.300	0.046		0.300	0.045	17,206
54	Protective services workers	0.100	0.008		0.100	0.011	3,969
61	Market-oriented skilled agricultural workers	0.182	0.027		0.182	0.051	1,325
62	Market-oriented skilled forestry, fishery and hunting workers	0.179	0.000		0.179	0.045	22
63	Subsistence farmers, fishers, hunters and gatherers	0.000	0.375		0.000	0.375	8
71	Building and related trades workers	0.027	0.008		0.027	0.016	5,208
72	Metal, machinery and related trades workers	0.000	0.008		0.000	0.013	3,982
73	Handicraft and printing workers	0.138	0.118		0.138	0.129	666
74	Electrical and electronic trades workers	0.000	0.022		0.000	0.042	3,424
75	Food, wood, garment and other craft workers	0.073	0.122		0.073	0.109	2,304
81	Stationary plant and machine operators	0.000	0.019		0.000	0.017	2,407
82	Assemblers	0.000	0.012		0.000	0.015	1,166
83	Drivers and mobile plant operators	0.237	0.006		0.237	0.006	8,480
91	Cleaners and helpers	0.000	0.004		0.000	0.003	3,474
92	Agricultural, forestry and fishery labourers	0.000	0.007		0.000	0.016	192
93	Labourers in mining, construction, manufacturing and transport	0.120	0.006		0.120	0.008	2,659
94	Food preparation assistants	0.000	0.013		0.000	0.011	649
95	Street and related sales and service workers	0.000	0.000		0.000	0.043	23
96	Refuse workers and other elementary workers	0.250	0.022		0.250	0.024	493

Ext.: the [Dingel and Neiman \(2020\)](#) teleworkability score at the two-digit ISCO-08 level, via the crosswalk described in Section 3.2. Real. 22–23: share of employed respondents (men and women, all ages in the extract) who usually work from home in 2022–23, the calibration’s benchmark; Swap: the calibration replaced the external score with it. Calib.: the score the hours DDD uses. Ref.-wk 21–23: share who worked from home in the survey reference week over 2021–23, men and women pooled, with *N* the pooled employed sample carrying that item. Codes 1–3 (armed forces) carry no external score and are not in the sample.